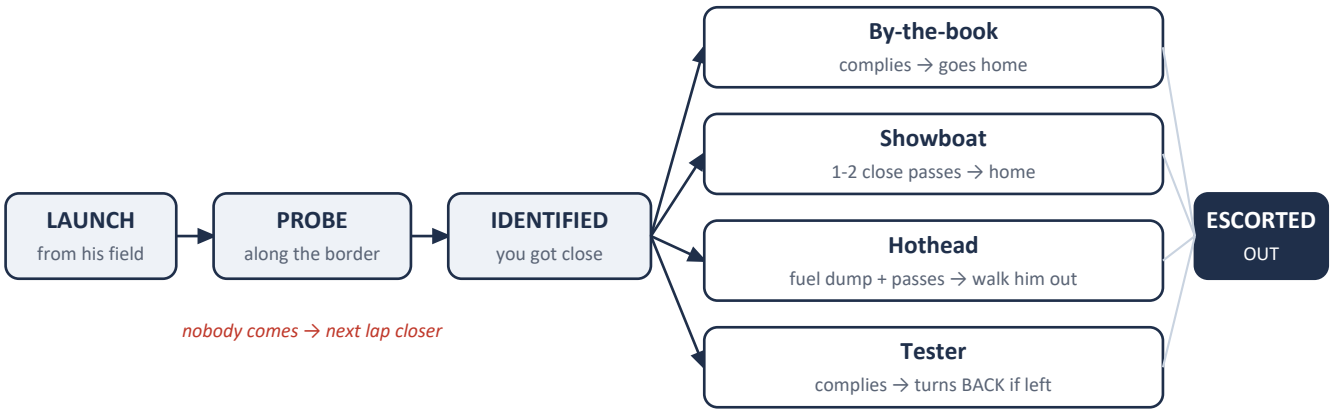
DAWS

Dynamic Air Warfare System

User Guide · v1.0.0-beta

> DAWS is a mission script for DCS World. Fighters launch from their own fields and **probe your border**: they fly along it, never cross it, and time how long you take to respond. When you intercept them, each one reacts according to the **pilot** it drew. One complies and goes home. Another dumps fuel across your windscreen and won't leave until you walk him out.



What you get	
No Lua needed	Set everything up in the Mission Editor with group names and flags.
Four pilot types	By-the-book, showboat, hothead, tester. Pick them, or let DAWS pick at random.
Real borders	The real Caucasus borders are built in. On other maps, draw your own zones.
Safe by design	Never crosses a border, never shoots. Close passes keep at least 100 m and abort inside 50 m.
Multiplayer	Runs on the server. Players need nothing installed.

Contents: 1 The pilots · 2 Setup in five steps · 3 Choosing how pilots act · 4 Flags · 5 Flying against it · 6 Recipes · 7 The config file · 8 Troubleshooting

1 The pilots

Every probe is flown by one of four pilot types. The type decides how close he pushes, what he does once you've identified him, and how hard he is to get rid of.

Pilot	Pushes	Once identified	Getting rid of him
By-the-book	as drawn	Complies within a minute and turns for home.	Easy: just escort him out.
Showboat	1 step closer	Comes over and does 1–2 close passes on you.	Goes home after his passes.
Hothead	2 steps closer	Opens with a **fuel dump** across your nose, then 2–4 passes: nose crossings, belly check, over the canopy, sits in your six.	Goes back to his probe track. Sit on his wing ~2.5 minutes and he gives up. Leave him alone and he may turn back.
Tester	1 step closer	Complies quickly, no passes.	**Always** turns back if nobody follows him out.

Three more behaviours apply to every pilot:

- **Unanswered probes push closer.** If nobody comes out, he runs the leg again a step (4 NM) closer, up to three laps. He never gets closer than 2 NM to the border.
- **Lanes remember.** A lane nobody defended starts its next probe deeper. A quick intercept and escort pushes it back out.
- **Shoot at him and he runs.** He breaks off and heads home at full speed. Nobody on the probing side ever fires.

Close passes

Pass	What you see
Fuel dump	Crosses ~260 m ahead, slightly high, trailing white smoke you fly through. Needs a smoke pod.
Nose cross	Slides across your nose ~400 m ahead.
Wing look / close wing	Sits on your wing at ~200 m, or tucks in to ~100 m.
Belly check	Slides underneath you at ~130 m.
Canopy pass	Passes ~150 m over your canopy.
Sit six	Sits in your six at ~450 m, gun range. He doesn't shoot.

Safety is automatic. No pass, and no path between two passes, comes inside 100 m. If anything gets inside 50 m he aborts and breaks away. Passes pause while you turn hard or fly below 1,500 m above the ground. There are no passes near a border or over land.

2 Setup in five steps

Everything below is done in the Mission Editor. The fastest way to learn it is to open **missions/DAWS_Starter_Caucasus_v1.0.miz** from the zip, where every step is already done.

Step 1 — Load the script

Trigger	Condition	Action
ONCE	TIME MORE 1	DO SCRIPT FILE → `DAWS_borders_caucasus.lua` (Caucasus only)
ONCE	TIME MORE 2	DO SCRIPT FILE → `DAWS.lua` (always last)

Use DO SCRIPT FILE, not DO SCRIPT with the text pasted in. You don't need to edit MissionScripting.lua.

Step 2 — Place templates (who flies the probes)

A template is an aircraft group DAWS copies every time it launches a probe. The original never flies.

- **Side:** the probing side (red by default).
- **Late activation:** ON.
- **Start:** Takeoff from runway, at his home field. He takes off, probes and lands back there.
- **Name:** must start with `DAWS`, e.g. `DAWS Su-27 Sochi`.
- **ROE:** nothing to set. DAWS puts every probe on weapon hold itself.
- **For the fuel dump:** put **Smoke Generator – white** on a pylon. The Su-27, Su-33, MiG-29, J-11A, Su-25 and L-39 can carry one; the MiG-31 can't, so it never dumps fuel.
- **Size:** one aircraft is enough. DAWS sends singles or pairs at random (see the #single / #pair tags).

Step 3 — Draw lanes (where they probe)

A lane is the route a probe flies along the border. Draw it with any late-activated group: it never spawns, it only marks the route.

- **Name:** `DAWS LANE GAGRA` (any name after DAWS LANE). The name is also used in its flag: `DAWS_LAUNCH_GAGRA`.
- **Waypoints:** two or more, drawn **parallel to the border and outside it**, about 10–15 NM off. DAWS pushes the route closer on its own when nobody answers.
- **Over land?** Add `#land` to the name (e.g. `DAWS LANE ROKI #land`). There are no passes over land; you shadow him from your own side, so identifying him only needs 15 NM.

Step 4 — Borders (what they won't cross)

- **Caucasus:** nothing to do. Loading `DAWS_borders_caucasus.lua` gives the real borders: Georgia (including Abkhazia and South Ossetia) and its 12 NM sea, Turkey and its 12 NM sea, Armenia and Azerbaijan.
- **Any other map:** draw polygon trigger zones named `DAWS BORDER` (anything after it). Every one is off-limits. Draw a separate polygon for the territorial sea.

Step 5 — Fly

That's it. With no other settings, the first probe launches at about 10 minutes, then one every 15–25 minutes, two at most at a time. OVERLORD calls each new track to the defending side.

3 Choosing how pilots act

There are four ways to decide who flies a probe, from broadest to most precise. You can mix them: a flag value beats any tag, a template's tag beats its lane's tag, and any tag beats the random choice.

You want	Do this	Example
One template always acts a certain way	Tag the template's name	`DAWS Su-27 Sochi #hothead`
A template only flies certain lanes	Tag it with @ and the lane name	`DAWS MiG-29 Krymsk @POTI`
Everyone on a lane acts one way	Tag the lane's name	`DAWS LANE GAGRA #showboat`
This one launch, right now	Set a launch flag (section 4)	`DAWS_LAUNCH_GAGRA` = 4
Random	Do nothing	45% book · 30% showboat · 15% hothead · 10% tester

All the name tags

Tag	On	Effect
`#book` `#showboat` `#hothead` `#tester`	template or lane	Always this pilot type.
`@GAGRA` `@POTI` …	template	Flies only these lanes (you can list several). Without any @, he flies every lane.
`#nostunts`	template	Never does close passes (e.g. a bomber, or a well-behaved escort).
`#smoke`	template	Treat him as having a smoke pod (normally read from his pylons).
`#single` `#pair`	template	Always one aircraft / always two.
`#land` `#sea`	lane	Over land or over water (normally worked out from the terrain).

You can combine tags: `DAWS Su-33 Sochi #hothead @GAGRA #pair` is always a hothead pair on the GAGRA lane.

> Big aircraft never do close passes, whatever their tag. That covers the Tu-95/142, Tu-22M3, Tu-160, A-50, Il-76/78, An-26 and Su-24MR. They still probe, so a `DAWS Tu-95 #book` is a classic Bear run down the coast.

4 Flags: controlling DAWS from triggers

DAWS uses **named flags**: type the name straight into any Mission Editor flag field. Use them to launch probes from your own triggers or from F10 radio items, and to react to what happens.

Flags DAWS watches (you set them)

Flag	Set it to	Effect
DAWS_LAUNCH_<LANE>	1 – 5	Launch a probe on that lane now (spaces in the lane name become _). The value picks the pilot.
DAWS_LAUNCH	1 – 5	Same, on a random lane.
DAWS_PAUSE	1 / 0	1 stops the automatic launches, 0 starts them again. Flag launches still work while paused.

Value	1	2	3	4	5
Pilot	random	by-the-book	showboat	hothead	tester

DAWS sets the launch flag back to 0 once the probe is up, so the same trigger or radio item can be used again.

Example: a GM control panel in the F10 menu

Trigger	Condition	Actions
ONCE	TIME MORE 3	ADD RADIO ITEM “Hothead on GAGRA” → flag DAWS_LAUNCH_GAGRA, value 4 ADD RADIO ITEM “Showboat on POTI” → flag DAWS_LAUNCH_POTI, value 3 ADD RADIO ITEM “Pause probes” → flag DAWS_PAUSE, value 1

Flags DAWS counts up (you read them)

Each time something happens, DAWS adds 1 to its flag. Use them as conditions: **FLAG IS MORE** DAWS_ESCORTED_OUT 0 → MESSAGE TO ALL “Good escort”.

Flag	Counts	Flag	Counts
DAWS_LAUNCHED	a probe launched	DAWS_ESCORTED_OUT	escorted out past 30 NM
DAWS_PROBE	he reached his route	DAWS_WITHDREW	went home with no escort
DAWS_INTERCEPT	he was identified	DAWS_UNANSWERED	a lap nobody responded to
DAWS_SHOW	close passes started	DAWS_TURNBACK	he turned back toward the border
DAWS_FUEL_DUMP	a fuel-dump pass	DAWS_FIRED_ON	someone shot at a probe
DAWS_SHOW_ABORT	passes aborted (too close)	DAWS_LOST	a probe crashed or was shot down

The F10 GM menu

The defending side gets **F10 → DAWS (GM)** automatically, with these entries:

- **Status:** every probe out, with its pilot, state, response time, who identified it and who escorted it.
- **Launch a probe now:** pick a lane.

- **Next probe's personality:** pick the next pilot.
- **Pause / resume:** automatic launches.

5 Flying against it

To	Do this
Identify him	Get inside **2 NM**, within **1,000 m** of his altitude, for **10 seconds**. OVERLORD calls the ID. On land lanes, shadow him from your own side of the border within 15 NM.
Get a clean show	Fly **straight and level**. He formates on you and runs his passes. Hard turns pause the show; getting within 50 m of him makes him abort.
Walk out a hothead	After his passes he goes back to his probe track. Stay on his wing about 2.5 minutes and he gives up.
Escort him out	Stay within **3 NM** (15 on land) until he's **30 NM** from the border. You need 60 s of escort to be credited.
Stop a turnback	Follow him out. Leave him alone on the way home and a tester will come back, and a hothead might.

> **Weapons tight.** Any shot or hit on a probe sends him home at speed, and DAWS logs it (`DAWS_FIRED_ON`). If you have **AI** fighters on the defending side, set their ROE to weapon hold, or they'll shoot the probes down.

6 Recipes: “I want…”

I want…	How
Every jet out of one field to be a hothead	Template name `DAWS Su-27 Sochi #hothead`.
A hothead on the coast at exactly 20 minutes	Trigger ONCE · TIME MORE 1200 · SET FLAG VALUE `DAWS_LAUNCH_GAGRA` = 4.
Only the probes I launch, none at random	Trigger ONCE · TIME MORE 1 · SET FLAG VALUE `DAWS_PAUSE` = 1. Then launch with flags. (Or `auto = false` in a config.)
The GM to pick the pilot live	F10 → DAWS (GM) → Next probe's personality, then Launch a probe now.
A Bear run down the coast	Template `DAWS Tu-95 #book @GAGRA` (bombers never do passes).
One lane to be quiet and professional	Lane name `DAWS LANE ROKI #book`.
Escalation: calm at first, aggressive later	Lane tagged `#book`; at 45 min a trigger sets `DAWS_LAUNCH_GAGRA` = 4 three times, a few minutes apart.
A reward when players do it right	FLAG IS MORE `DAWS_ESCORTED_OUT` 2 → message / sound / next phase.
A consequence when they get it wrong	FLAG IS MORE `DAWS_FIRED_ON` 0 → your incident trigger. FLAG IS MORE `DAWS_UNANSWERED` 2 → “they're getting bolder”.
A hothead that's harder to get rid of, or a new pilot type	Config file (next section).

7 The config file (optional)

Tags and flags choose *which* pilot flies. The config file changes *how* pilots behave and how DAWS runs. Copy `DAWS_config_template.lua` from the zip, uncomment the lines you want, and load it between the borders file and `DAWS.lua`. Every option in it is commented with its default; these are the common ones.

Make hotheads harder to walk out, and change the odds

```
DAWS_CONFIG = {
  personalities = { hothead = { escort_need_s = 300 } },   -- 5 minutes on his wing instead of 2.5
  weights = { book = 30, showboat = 30, hothead = 30, tester = 10 },
}
```

A new pilot type

```
DAWS_CONFIG = {
  personalities = {
    cowboy = { label = "cowboy", depth = 2,           -- starts 2 steps closer
               stunts = { 3, 3 }, opens_with = "sit_six",
               pool = { "sit_six", "canopy_pass", "nose_cross" },
               escort_need_s = 60, turnback = 0.5 },  -- walk him out for 60 s; 50% he turns back
  },
  weights = { book = 40, showboat = 30, hothead = 10, tester = 10, cowboy = 10 },
}
```

Tag templates `#cowboy` to use it, or launch with flag value 6 (your own pilots are numbered from 6, alphabetically).

A timed script of probes

```
DAWS_CONFIG = {
  auto = false,                                    -- nothing at random
  schedule = {
    { at = 300,  lane = "GAGRA", personality = "book",    tag = "Act 1" },
    { at = 1500, lane = "GAGRA", personality = "hothead", tag = "Act 2" },
    { at = 2400, lane = "ROKI",  every = { 1200, 1800 } },   -- then every 20-30 min
  },
}
```

The tag shows in front of that probe's messages, e.g. [Act 2].

Other settings people change

Option	Default	What it does
`first_s`, `interval_s`	600, {900,1500}	When the first automatic probe comes, then how often (seconds).
`max_active`	2	Most automatic probes out at once.
`floor_nm`, `step_nm`	2, 4	Closest to the border he ever gets; how much closer each push goes.
`intercept_nm`, `intercept_s`	2, 10	What counts as identifying him.
`release_nm`, `escort_credit_s`	30, 60	What counts as escorting him out.
`awacs`	"OVERLORD"	The callsign that makes the calls.

Option	Default	What it does
`side`	"red"	Which side probes (set "blue" to flip it).
`roe`	4	The probing side's ROE: 4 = weapon hold, 3 = return fire.

8 Troubleshooting

DAWS writes every event to `Saved Games\DCS\Logs\dcs.log` as lines starting with `DAWS|`. Search the log for them first.

You see	What it means / what to do
Nothing happens at all	Search the log for `[DAWS] loaded` and `Mission script error`. Is DAWS.lua loaded **last**, with DO SCRIPT FILE?
`WARN|no templates` `WARN|no lanes`	Check the names (`DAWS …` / `DAWS LANE …`), late activation, and that templates are on the probing side.
`WARN|no borders defined`	Load the borders file (Caucasus), or draw `DAWS BORDER` zones.
`LANE_WARN|… only x NM off`	A lane waypoint is too close to the border. Move it further out.
`SPAWN_COUNTRY`	Normal. DAWS swapped to a country the game really puts on the probing side, so the jets come out the right colour.
`SHOW_STEP|…|timeout`	He didn't reach one pass position in 45 s. A few are normal. If every pass times out, send the log.
`STUNT_REJECTED`	A pass you wrote comes inside 100 m of the target, so DAWS won't fly it.
No fuel dump	The template has no white smoke pod, or the type can't carry one (MiG-31).

What's in the zip

File	What
`DAWS.lua`	The script. Load it last.
`DAWS_borders_caucasus.lua`	Real Caucasus borders (from Natural Earth, public domain).
`DAWS_config_template.lua`	Every option, commented. Optional.
`examples/DAWS_config_Caucasus.lua`	A complete config written in Lua instead of the Mission Editor.
`missions/DAWS_Starter_Caucasus_v1.0.miz`	Open in the Mission Editor: everything in this guide, set up.
`missions/DAWS_Demo_Caucasus_v1.3.miz`	Spectate: five intercepts with five different pilots, all AI.
`MANUAL.md`	The full technical reference, including the Lua scripting API.

> **Beta.** The full probe → intercept → passes → fuel dump → turnback → escort-out cycle has run in DCS. The formation steering for close passes is new in this version. If passes look wrong, send the `SHOW_STEP` lines from your dcs.log to github.com/ExpertFett/DAWS/issues.

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